

Solution

SOUND-04

Class 09 - Science

1. (a) (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv)

Explanation: (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv)

2.

(c) Only (i) is true.

Explanation: Only (i) is true.

3.

(c) $W = FS \cos \theta$ and $F = \text{Force}$

Explanation: Work is defined as a force acting upon an object to cause a displacement. Mathematically, work can be expressed by the following equation.

$$W = F \times S \cos \theta$$

Where W is work is done, S = displacement and F = Force

4. (a) the momentum of the system is constant

Explanation: The total momentum of the system just before the collision is the same as the total momentum just after the collision. So, the momentum of the system is constant.

5.

(c) a force of 1 N displaces a body by 1 m

Explanation: 1 joule(J) is the amount of work done when a force of 1 Newton(N) displaces an object 1 meter(m).

$$1J = 1N \times 1m \text{ or } 1J = 1Nm.$$

6. (a) Only energy transformation takes place

Explanation: The pendulum swings down and the potential energy is converted to kinetic energy. At the bottom of the swing, the potential energy is all gone (this is the lowest it goes) and it is all kinetic energy. This is also the fastest that it goes.

7.

(b) 3.6×10^6 joule

Explanation: One kilowatt-hour is equal to 3600000 joules:

$$1kWh = 3600000J = 3.6 \times 10^6J$$

8.

(c) the body moves along the direction of the applied force

Explanation: The work done is positive if the applied force is in the same direction as the direction of motion.

9.

(d) Only (ii) is true.

Explanation: The speed of the object is affected by the height from which it is released.

$$mgh = \frac{1}{2}mv^2 \text{ or } v = \sqrt{2gh} \text{ or } v \propto \sqrt{h}$$

So, heavier object will not move with a greater speed at Y.

Speed of the object at M:

$$\text{Loss in PE} = \text{KE}$$

$$mg(25 - 20) = \frac{1}{2}mv^2 \therefore v^2 = 100 \text{ or } v = 10 \text{ ms}^{-1}$$

10.

(d) 27

$$\text{Explanation: Power} = \frac{\text{Energy consumed}}{\text{Time}}$$

$$\text{Given power} = 60W = 0.06kW$$

$$\text{Therefore, For one-day energy consumed, } 0.06kW = \frac{\text{Energy consumed}}{5 \text{ hr}}$$

$$0.30kWh = \text{Energy consumed}$$

0.30 unit one day.

$$\text{Now for 30 days,} = 0.30 \times 30 = 9 \text{ UNIT}$$

Given for 1 unit cost = ₹ 3

Hence cost for 9 unit = $9 \times 3 = ₹ 27$

11.

(b) $S_1 = S_2$

Explanation: Let m_1, m_2 be the masses and u_1, u_2 be the initial velocities of the truck and car respectively.

As KE of both are equal.

$$\text{So, } \frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_2 u_2^2 \text{ or } \frac{m_1}{m_2} = \frac{u_2^2}{u_1^2}$$

As equal retarding force is applied on both then

$$0 = u_1^2 - \frac{2fS_1}{m_1} \Rightarrow S_1 = \frac{m_1 u_1^2}{2f}$$

$$\text{and } 0 = u_2^2 - \frac{2fS_2}{m_2} \Rightarrow S_2 = \frac{m_2 u_2^2}{2f}$$

$$\text{So, } \frac{S_1}{S_2} = \frac{m_1}{m_2} \frac{u_1^2}{u_2^2} \therefore \frac{S_1}{S_2} = \frac{1}{1} \text{ or } S_1 = S_2$$

12.

(c) $mg \frac{h}{2}$

Explanation:

If you consider gravitational field to be constant till height 'h' then we know that Center of the mass of tank lies $\frac{h}{2}$ height above ground level.

Now potential energy will be $= m \times g \times \left(\frac{h}{2}\right)$.

13.

(d) Work and energy

Explanation: The SI unit of work is joule and is denoted as 'J', which is named after an English physicist James Prescott Joule.

The 1 joule of work done is equal to $1\text{N} \times 1\text{m}$. Or, $1\text{ joule} = \text{Nm}$

Energy is capable of doing work, therefore, the SI unit of energy is the same as of work.

Thus, the SI unit of energy is joule and is denoted by 'J'. A larger unit of energy is kilojoule and is denoted by kJ.

14.

(b) 20m s^{-1}

Explanation: Change in gravitational potential energy of the skier, $E_p = mgh = 50 \times 10 \times [60 - 20] = 20000\text{ J} = 2 \times 10^4\text{ J}$

Kinetic energy, $E_K = 50\%$ of $E_p = \frac{1}{2} \times 2 \times 10^4\text{ J} = 1 \times 10^4\text{ J}$

$$E_K = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2E_K}{m}} = \sqrt{\frac{2 \times 1 \times 10^4}{50}} = 20\text{ m s}^{-1}$$

15.

(c) remains constant

Explanation: Since, total energy of the system is always conserved, so when a body falls freely towards the earth, then its total energy remains constant i.e., the sum of the potential energy and kinetic energy of the body would be same at all points.

16.

(a) 746 W

Explanation: 1 HP = 550 Foot Pound per second.

1 Pound = 4.44822 meters.

1 foot is 0.3048 meters.

1 Hp = $550 \times 4.44822 \times 0.3048$ meters

= 745.69 Nm/s = 746 Watts.

17.

(b) there is no component of force, parallel to the direction of motion

Explanation: Work is done whenever a force or a component of a force results in a displacement. No component of the force is acting in the direction of motion.

18.

(c) 2.68 h.p

Explanation: Velocity = $36\text{km/h} = 10\text{m/s}$

$$w = f \times s = 2000 \times 10 = 2000 \text{ J}$$

$$P = \frac{W}{t} = \frac{2000}{1} = 2000 \text{ W}$$

$$746 \text{ W} = 1 \text{ hp}$$

$$\text{So, } 2000 \text{ W} = \frac{2000}{746} = 2.68 \text{ h.p.}$$

19.

(b) Statement 1 is true but statement 2 is false

Explanation: Potential energy (U) = $\frac{1}{2}kx^2$ i.e. $U \propto x^2$

This is the equation of parabola, so graph between U and x is a parabola, not straight line.

20.

(c) energy

Explanation: Commercial unit of energy is kilowatt hour (kWh).

21.

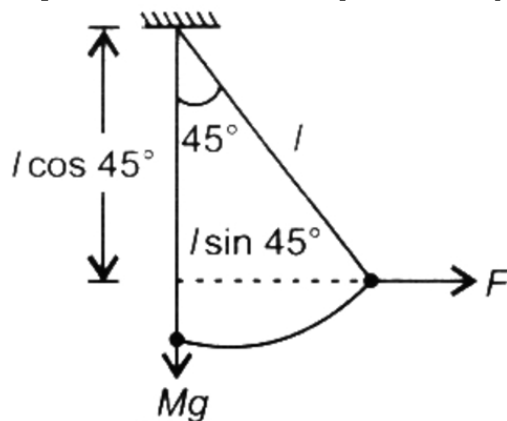
(c) Fv

Explanation: Power = Force $\times$ velocity = $F \times v$

22.

(c) $Mg(\sqrt{2} - 1)$

Explanation: Work done in displacement is equal to gain in potential energy.



$$\text{Work done} = F \times l \sin 45^\circ = \frac{Fl}{\sqrt{2}}$$

Gain in potential energy

$$= Mgl(1 - \cos 45^\circ)$$

$$= MgI \left(1 - \frac{1}{\sqrt{2}} \right) \therefore \frac{Fl}{\sqrt{2}} = \frac{MgI(\sqrt{2}-1)}{\sqrt{2}}$$

$$\text{or } F = Mg(\sqrt{2} - 1)$$

23. (a) joule

Explanation: SI unit of energy is the same as of work. The SI unit of energy is joule and is denoted by 'J'.

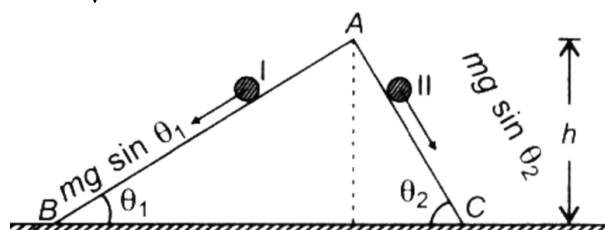
24.

(b) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.

Explanation: AB and AC are two smooth planes inclined to the horizontal at different angles.

As height of both the planes are same, therefore, both the stones will reach the bottom with same speed.

$$\therefore v = \sqrt{2gh} \therefore v_1 = v_2$$



From figure, acceleration, $a_1 = g \sin \theta_1$

acceleration, $a_2 = g \sin \theta_2$

As $\theta_2 > \theta_1$, so $a_2 > a_1$

Using, $v = u + at$

$$v_1 = u_1 + a_1 t_1 ; v_1 = a_1 t_1$$

$$t_1 = \frac{v_1}{a_1}$$

$$\text{Similarly, } t_2 = \frac{v_2}{a_2} \because a_2 > a_1 \text{ So, } t_2 < t_1$$

Hence, stone II takes less time to reach the bottom.

25.

(b) Potential energy

Explanation: The energy possessed by an object because of its position is called potential energy. For example; when a stone is kept at a height, it possesses some energy because of its height. Because of this potential energy, the object kept at a height falls over the ground.

26.

(c) 1.3

Explanation: Force = $m g = 500 \text{ kg} \times 9.8 \text{ m/s}^2 = 4900 \text{ N}$

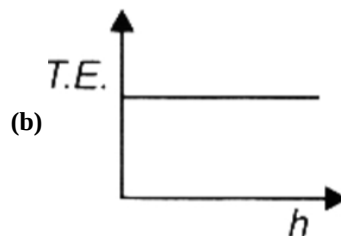
Velocity = 0.2 m/s

Now, power = Force $\times$ velocity = $4900 \times 0.2 = 980 \text{ W}$

Now 1 H.P. = 746 watt

So, $980 \text{ W} = \frac{980}{746} = 1.3 \text{ hp}$

27.



Explanation: Total energy remains constant.

28. **(a)** J/s

Explanation: Power is measured in energy (joules) divided by time. The SI unit of power is the watt (W) or joule per second (J/s). Power is a scalar quantity, it has no direction.

29. **(a)** Positive

Explanation: Work = $\frac{1}{2} m(v^2 - u^2)$

If the object starts moving from the state of rest, therefore, initial velocity (u) will be equal to zero.

Therefore, equation (iii) can be written as

$$W = \frac{1}{2} m (v^2 - 0^2)$$

$$W = \frac{1}{2} m v^2$$

W = Positive kinetic energy

30.

(d) less than cone

Explanation: $P \propto \frac{1}{A}$

The pressure is inversely proportional to the area and $A_{\text{Cylinder}} > A_{\text{Cone}}$

31.

(d) remains the same

Explanation: The kinetic energy changes into the potential energy when stone rises up. The potential energy then changes back to the kinetic energy, when a stone falls down. A boy throws a ball vertically upwards. As the ball rises its total energy remains the same.

32.

(d) 1 joule

Explanation: 1 joule(J) is the amount of work done when a force of 1 Newton(N) displaces an object 1 meter(m).

$$1\text{J} = 1\text{N} \times 1\text{m} \text{ or } 1\text{J} = 1\text{Nm.}$$

33. **(a)** 245 J

Explanation: P.E. = mgh

$$m = 5\text{kg } h = 5\text{m } g = 9.8\text{m/s}^2$$

$$\text{P.E.} = 5 \times 9.8 \times 5$$

$$\text{P.E.} = 25 \times 9.8$$

$$\text{P.E.} = 245 \text{ J}$$

34.

(b) He does zero work against the force of gravity

Explanation: He does zero work against the force of gravity because no displacement is done.

35.

(b) potential energy

Explanation:

Potential energy is the energy of a body due to its height, hence any object that is not at the center of the earth is under a gravitational pull and has an energy that can enable it to travel if unhindered towards the center. Hence all the objects on earth that are at rest, have gravitational potential energy.

36.

(d) 40 W

Explanation: In series 40W bulb glow brighter as

$$P = VI \text{ (P – power)}$$

$$P = I^2R \text{ (V= IR)}$$

Therefore power is directly proportional to resistance. The resistance of the 40W bulb is higher.

So, it will have more power and it will glow brighter.

37.

(d) thrust is same but effects are different

Explanation: As the contact area changes, the effect of thrust changes.

38.

(a) 1960 W

Explanation: $m = 240 \text{ kg}$, $g = 9.8 \text{ m/s}^2$

$$\text{Force} = \text{weight} = mg = 240(9.8) = 2352 \text{ N}$$

displacement = 2.5m and $t = 3 \text{ second}$

$$\text{Power} = \frac{\text{force} \times \text{displacement}}{\text{time}} = \frac{2352 \times 2.5}{3} = \frac{5880}{3} = 1960\text{W.}$$

39.

(d) half potential and half kinetic energy

Explanation: As we know that when a body is at height h , it has

total energy = KE + PE

and at height ' h ', velocity of body is zero

SO, KE = 0 and PE = mgh

Now, at height $\frac{h}{2}$

$$\text{its PE} = mg \frac{h}{2} = \frac{mgh}{2} \dots\dots\dots (i)$$

$$\text{and KE} = \frac{1}{2} mv^2$$

Where, velocity can be determined by equation of motion

$$v^2 = u^2 + 2gh$$

$$\text{So, } v^2 = \frac{2gh}{2} \Rightarrow v = \sqrt{gh}$$

$$\text{So, KE} = \frac{1}{2} m(\sqrt{gh})^2 \dots\dots\dots (ii)$$

So, from equation (i) and (ii) body at $(\frac{h}{2})$ possesses half potential energy and half kinetic energy.

40.

(a) $E = Pt$

Explanation: Energy = Power $\times$ time

i.e. $E = \text{Watt} \times \text{time}$

When we say we have consumed 1 unit of electricity, that means we have consumed 1kWhr of electrical energy.

1 Unit = kilo watt-hour.

i.e. 1unit = 1kWhr

41. (c) kilowatt
Explanation: Since energy is capacity of doing work, therefore, the SI unit of energy is same as of work.
42. (c) It depends upon the area of contact for the same magnitude of the force:
Explanation: $P = \frac{F}{A}$, For the same force $P \propto \frac{1}{A}$
43. (a) kinetic energy increases 4 times
Explanation: K.E. = $\frac{1}{2} mv^2$
 If speed double so, $v = 2v$
 $K.E. = \frac{1}{2} m(2v)^2 = \frac{1}{2} m(4v^2) = 4(\frac{1}{2} mv^2)$
44. (c) K.E. of the heavier is more than that of the lighter.
Explanation: P.E. of object1 = mgh
 P.E. of object 2 = 2mgh
 On reaching the ground the P.E. of both the objects gets completely transformed into K.E.
 Therefore K.E. of object 2 > K.E. of object 1.
45. (c) zero
Explanation: The work done is zero because between the road and girl the angle made is 90 so the value of cos 90 is zero so when we use formula $w = f s \cos 90$ the answer will zero.
46. (d) 1000w
Explanation: Force = 100 N
 Velocity = 36 k m/h = $36 \times \frac{5}{18} = 10$ m/s
 Power = Force $\times$ Velocity Power = 100 $\times$ 10
 Power = 1000 Watt
 Therefore the power of rickshaw puller = 1000 Watt
47. (a) potential energy
Explanation: Water stored in a dam possesses potential energy. And once the water is falling it has kinetic energy.
48. (c) half
Explanation: On removing the disc, weight reduces to half.
49. (c) 12.5 m
Explanation: We know, if a body is thrown up vertically its KE goes on decreasing and PE goes on increasing. There is conversion of KE into PE. When kinetic energy becomes half of original value, the other half is converted into potential energy.
 Therefore, $\frac{490}{2} = 2 \times 9.8 \times h$
 or $h = \frac{490}{2 \times 2 \times 9.8}$ or $h = 12.5$
 Thus, the required height is 12.5 m.
50. (b) Coolie B
Explanation: Power = $\frac{\text{Work}}{\text{time}}$
 $\therefore$ Work done by both of them is same.
 Hence, coolie who have taken less time has greater power i.e., coolie B.
51. (a) 6 m
Explanation: Energy = mgh = $m \times 10 \times 10 = 100m$ Joule.
 Energy is reduced by 40% then the remaining energy is 60m Joule.
 Therefore $60 m = m \times 10 \times h$
 or $h = 6$ m

52. **(d)** 375 W
Explanation: Weight of the boy, $mg = 50 \text{ kg} \times 10 \text{ m s}^{-2} = 500 \text{ N}$
 Height of the staircase, $h = 45 \times \frac{15}{100} \text{ m} = 6.75 \text{ m}$
 Time taken to climb, $t = 9 \text{ s}$

$$\text{Power} = \frac{\text{Work done}}{\text{time taken}}$$

$$P = \frac{mgh}{t} = \frac{500 \text{ N} \times 6.75 \text{ m}}{9 \text{ s}} = 375 \text{ W.}$$
 Power is 375 W.
53. **(c)** 150 W
Explanation: Here, mass, $m = 50 \text{ kg}$
 Height of the staircase, $h = 120 \times \frac{25}{100} = 30 \text{ m}$
 Time taken to climb = 1 min 40 s = 100 s
 We know, Power, $P = \frac{\text{Work done}}{\text{Time taken}} = \frac{mgh}{t}$

$$= \frac{50 \times 10 \times 30}{100} = 150 \text{ W}$$
54. **(a)** 10 s
Explanation: Due to conservation of energy,
 input energy = output energy
 Power $\times$ time = increase in kinetic energy

$$P \times t = \frac{1}{2} [mv^2 - mu^2]$$

$$(20000)(t) = \frac{1}{2} [(800)(30)^2 - (800)(20)^2] \text{ or } t = 10 \text{ s}$$
55. **(b)** zero
Explanation: When a body is whirled in a circle the work done by the centripetal force is zero Because work is done is defined as the distance an object is moved in the direction of the force applied and since the centripetal force acts perpendicular to the velocity (direction of motion) the work done is 0. Work done by the centripetal force is zero only when the radial position does not change.
56. **(c)** 9 times
Explanation: K.E. is proportional to V^2 .
 So, when speed is tripled, KE. is nine-time.
57. **(a)** decreased by 3.5 times
Explanation: $P_1 = \frac{F_1}{A_1} = \frac{M_1 g}{A_1} = \frac{5 \times 9.8}{0.30 \times 0.15} = 1.09 \times 10^3 \text{ N/m}^2$

$$P_2 = \frac{F_2}{A_2} = \frac{(M_1 + M_2)g}{\pi r^2} = \frac{(5 + 1) \times 9.8}{0.0154} = \frac{58.8}{0.0154} = 3.82 \times 10^3 \text{ N/m}^2$$

$$\frac{P_2}{P_1} = \frac{3.82 \times 10^3}{1.09 \times 10^3} = 3.5$$
58. **(d)** erg
Explanation: The unit of work in the cgs system is dyne cm or erg.
59. **(a)** 80 times
Explanation: Total work = $p \times t = 120 \text{ J}$,
 Number times heartbeat in 1 min. = total work done / work done in each beat

$$= \frac{120}{1.5} = 80 \text{ times}$$
60. **(c)** a person is lying on the ground
Explanation: The potential energy of your body is least when you are lying down on the ground. An object can store energy as a result of its position. The stored energy of position is called potential energy.
61. **(d)** In moving from C to D, work done by the force on the body is positive

Explanation: In moving from C to D, slope of graph is negative which means acceleration (force) is negative and displacement is also negative as the area under curve CD lies below the time axis. Hence, work done by the force on the body is positive.

62.

(d) 0.102 m

Explanation: P.E. = mgh

$$m = 1 \text{ kg P.E.} = 1 \text{ J } g = 9.8 \text{ m/s}^2$$

$$1 = 1 \times 9.8 \times h$$

$$h = \frac{1}{9.8} = 0.102 \text{ m}$$

63.

(c) acceleration

Explanation: When both spheres are dropped simultaneously from a cliff, they have the same acceleration. Because during free-fall acceleration of the body becomes equals to ($g = 9.8 \text{ m/s}^2$) and 'g' depends on the mass of earth and radius of the earth.

64.

(d) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

Explanation: While carrying a bucket, angle between vectors F and S becomes 90° .

$$\text{So, } W = \vec{F} \cdot \vec{S} = FS \cos \theta = FS \cos 90^\circ = 0$$

Hence, work done is zero.

65.

(a) zero

Explanation: The object moves in a circular path. In this case, the centripetal force acts on the object at right angles to the direction of motion of an object. So, the work done by the object in a circular path is zero.

66.

(b) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

Explanation: Relation between momentum and KE (E_K) is

$$p = \sqrt{2mE_K}$$

If mass of body decreases, then KE should increase accordingly to maintain constant momentum.

$$\text{Also, } E_K = \frac{1}{2}mv^2 \text{ or } E_K \propto v^2$$

67. (a) 52,02,624 joule

Explanation: Mass $m = 60 \text{ kg}$

height $h = 8848 \text{ m}$

$$g = 9.8 \text{ m/s}^2$$

$$\text{Potential energy } U = mgh = (60 \times 9.8 \times 8848) = 52,02,624 \text{ J}$$

68. (a) 180°

Explanation: When force is applied in opposite direction of displacement, the work done is considered as negative and the angle between the force and displacement is 180° .

69. (a) joule

Explanation: The SI unit of work is joule and is denoted as 'J', which is named after an English physicist James Prescott Joule.

The 1 joule of work done is equal to $1 \text{ N} \times 1 \text{ m}$. or, $1 \text{ joule} = \text{Nm}$.

70. (a) Negative

Explanation: When force is applied in the direction of displacement, the work done is considered as positive.

$$\text{i.e. } W = F \times s$$

When force is applied in the opposite direction of displacement, the work done is considered negative.

$$\text{i.e. } W = -F \times s = -Fs$$

71. (a) 10 m

Explanation: P.E. = mgh

$$m = 50 \text{ kg P.E.} = 4900 \text{ J } g = 9.8 \text{ m/s}^2$$

$$4900 = 50 \times 9.8 \times h$$

$$4900 = 490 \times h$$

$$h = \frac{4900}{490} = 10 \text{ m}$$

72. (a) momentum is conserved

Explanation: When a cracker bursts, its pieces move out in all directions. We explain it with the help of the law of conservation of momentum. In this case, initially, the momentum of the cracker is zero. Therefore, when it bursts, its pieces move out in different directions with some specific momentum. However, if we add up the momentum of these various pieces, the sum of the momenta will be zero.

73.

(d) opposite parallel faces of the cuboid exerted some pressure on the sand

Explanation: Since force per unit area remains the same, so the same conclusion is drawn by both the students.

74.

(c) Electrical signal/energy

Explanation: Microphones are loudspeakers in reverse, it converts sound (mechanical) energy to electrical energy.

75.

(c) position

Explanation: The energy possessed by an object because of its position is called potential energy.